BEING AN OPEN MORPHISM IS FPPF LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'Hence we win.' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The general criterion for a property of morphisms to be fppf local on the source. Same chapter, Lemma lemma-properties-morphisms-local-source, invoked at proof lines 7077-7078. Reduces the claim to four clauses.
- Openness is preserved under composition and flat morphisms of finite presentation are open. Morphisms, Lemmas morphisms-lemma-composition-open and morphisms-lemma-fppf-open, invoked at proof lines 7073-7076. Supplies clauses (1)-(3).
- The fpqc quotient topology criterion. Morphisms, Lemma morphisms-lemma-fpqc-quotient-topology, invoked at proof lines 7079-7080. Supplies clause (4).

Source reference. Original lemma 036H: sources/stacks-descent/pinned-original-source.tex, source lines 6920–6920. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02V2: sources/stacks-chow/tags.txt, source lines 3753–3753. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02JY: sources/stacks-chow/tags.txt, source lines 3353–3353. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is open" is fppf local on the source.*

Proof. Being an open morphism is clearly Zariski local on the source and the target. It is a property which is preserved under composition, see Morphisms, Lemma 02V2, and a flat morphism of finite presentation is open, see Morphisms, Lemma 01UA This proves (1), (2) and (3) of Lemma 036H. The final condition (4) follows from Morphisms, Lemma 02JY. Hence we win. □